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PI Name:	Hyde, Truell Ph.D.		
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PI Email:	Truell_Hyde@baylor.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	254-710-6717
Organization Name:	Baylor University		
PI Address 1:	Center for Astrophysics, Space Physics, and Engineering Research (CASPER)		
PI Address 2:	1 Bear Pl Unit 97283		
PI Web Page:			
City:	Waco	**State:**	TX
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Contact Monitor:	Callas, John	**Contact Phone:**	
Contact Email:	john.l.callas@jpl.nasa.gov		
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COI Name (Institution):	Matthews, Lorin Ph.D. (Baylor University) Hartmann, Peter Ph.D. (Baylor University) Petrov, Oleg Ph.D. (Joint Institute for High Temperatures, Russian Academy of Sciences) Rosenberg, Marlene Ph.D. (University Of California, San Diego)		
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Task Description:	Complex plasma consists of ions, electrons, neutral gas, and solid microparticles referred to as dust. Dust grains immersed in plasma become negatively charged and interact with one another as well as plasma components through electromagnetic interactions. In particular, dust grains experience a plasma drag force, altering the trajectories of ions in the plasma, and participate in collective plasma interactions, such as waves and instabilities. An interesting aspect of complex plasma is its ability to self-organize into dust liquids, 2D and 3D dust crystal lattices, and 1D dust chains, which are aligned with electric fields in the plasma. This feature provides researchers the ability to examine classical collective phenomena on size and time scales allowing for direct observation (at the kinetic level) of fundamental microscopic and macroscopic processes, as the dust can easily be imaged with high-speed cameras. In many environments, positive ions in the plasma have a directed flow with respect to the negatively charged dust grains. The resulting interaction between the dust and flowing plasma creates an ion wakefield downstream from the dust particles, with the positive space region modifying the interaction between the charged grains and contributing to the observed dynamics and particle structure. Since the ion wakefield is weak, the effects are usually masked by gravity for terrestrial experiments. This international research collaboration employs the Plasma Krystall-4 (PK-4) on the International Space Station (ISS) to conduct experiments in microgravity which investigate the underlying physics behind self-ordering of interacting complex plasma dust particles into field-aligned chains. Scientific Goals / Objectives of the Research -- The goal of the proposed research project is to investigate the underlying physics behind self-ordering of interacting complex plasma dust particles into field-aligned chains in a microgravity environment. This goal will be accomplished by pursuing the following specific objectives: 1. Establishing the complex plasma parameters required for the formation and stability of extended field-aligned dust particle chains, 2. Measuring the grain-grain potential along and perpendicular to the ion flow direction, and 3. Studying interchain interactions and their role in the structural properties of an ordered complex plasma system. Significance of the Research -- This research will be used to develop comprehensive theories on interparticle potential and grain-plasma interactions which determine the microscopic and macroscopic properties of complex plasma, including external forces acting on the particles, the fundamental interactions between the particles, thermodynamics, and self-organization of the system. The results will directly impact research in astrophysics and planetary science (dust in protoplanetary disks; charged dust on the surface of airless bodies, such as the moon and asteroids; planetary ring systems), atmospheric physics (noctilucent clouds), fusion research (dust contamination, operating stability), materials physics (soft matter, metastable states, complex fluids, granular matter), and advanced manufacturing (semiconductor, etching, and deposition systems).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Impact on Principle Discipline – Baylor – An analog PK-4 BU has been developed and is now operational. It is currently one of two such devices in the world and allows the ability for researchers to identify new parameter regimes and experimental plans which cannot be developed (or tested) on the current PK-4 ground unit. A new code has been developed to model the interactions of dust with flowing ions. This allows the dust charging and dynamics to be self-consistently calculated, including the effects of ion wakefield on the dust-dust interactions and charging. While many features to be implemented within the code are specific to the PK4 discharge, the code is robust and can be applied to many different dusty plasma experiments by changing the boundary conditions governing the ion flow and external confining potentials. The code allows users to explore the parameter space of experimental conditions to determine those which are favorable for a particular dust configuration, such as one-dimensional strings. Institute for Solid State Physics and Optics, Wigner Research Centre for Physics - The results of the particle-in-cell/Monte Carlo collision (PIC/MCC) simulations of the PK-4 gas discharge predicting the presence of fast ionization waves in both the stationary dc discharge and during the polarity switching have now been experimentally verified. These rapid features remained undetected in the PK-4 (ISS) experiment due to lack of fast diagnostics but are apparently of key importance for the formation of the observed dust particle chains. These findings should provide the necessary foundation for design of future experimental campaigns. The field of complex plasmas is now considered the plasma state of Soft Matter. As such, these results will push research in this field forward. Impact on the Development of Human Resources Baylor - To date, four undergraduate students (with two of these members of under-represented minorities in physics) have been involved in the PK-4 project at Baylor. As part of this project, students have gained experience in both experimental and numerical skills as well as practice in analyzing data, writing and presenting papers. The numerical students have additionally gained experience in testing and developing code, learned how to program using CUDA, and sharpened their skills in C++ and Matlab. Each of these programming languages is listed among the top twenty programming languages for 2017 by the Institute of Electrical and Electronics Engineers (IEEE). Impact on Professional Development – Baylor – Three graduate students and one undergraduate student have been involved in the development and testing of the PK-4 BU device over the past year. As part of this process, the students learned basic experimental and computer programming skills as well as high voltage, vacuum, and laser safety. Additionally, two undergraduate students and two graduate students have been involved in the development and testing of the computer code over the past year. The students learned how to program using CUDA in addition to sharpening their skills programming in C++ and Matlab. The above led to undergraduate Evie Guay giving three oral presentations on her research, including at the Conference on Undergraduate Women in Physics (CUWIP) held in Corpus Christi, TX, January 18, 2019. Dustin Sanford presented his work at the XXX IUPAP (International Union of Pure and Applied Physics) Conference on Computational Physics, CCP2018, University of California, Davis, July 29 – August 2, 2018 and at the 60th Annual Meeting of the American Physical Society Division of Plasma Physics (APS-DPP), Portland, OR, November 5-9, 2018, where he won an award for outstanding undergraduate research. The students also presented their results as part of the Baylor Physics Department colloquium series in September, 2018, and will make poster presentations during URSA Scholars Week, March 2019. Graduate students Zhiyue Ding, Mudi Chen, and Khandaker Sharm Ashrafi made presentations on their work at the 15th Dusty Plasma Workshop, Baltimore, Maryland, May 29-June 1, 2018 and the 60th Annual Meeting of the APS-DPP, Portland, OR, November 5-9, 2018.

Additionally, over the past year CASPER students (both graduate and undergraduate) have had the opportunity to attend meetings with the Principal Investigator (PI) and Co-Is from all over the world (Peter Hartmann, Hubertus Thomas, Vladimir Nosenko, Marlene Rosenberg) either at Baylor or at various conferences attended by the PI and students. Additional meetings with Ed Thomas (Auburn) and Jeremiah Williams (Wittenberg) and their students have also been held both at conferences and at Baylor.

Impact on Physical / Numerical Resources –

Baylor & Institute for Solid State Physics and Optics, Wigner Research Centre for Physics - Motivated by the new insights provided by both the numerical simulation of the gas discharge plasma (Wigner Research Centre for Physics in Budapest, Hungary) and the PK-4 BU device (CASPER HIDPL (Hypervelocity Impacts and Dusty Plasmas Lab)) laboratory at Baylor University) experimental verification of numerical results are now underway. This should help in the design of future experimental campaigns on the PK-4 facility at the ISS and the latest version of the Plasma Krystal family. Additionally, the PK-4 BU is a new tool which is already proving useful for testing potential experiments on the PK-4 on the ISS. The PK-4 BU is being used by multiple US research groups (including graduate students) and should prove useful for years to come. Finally, the IonWake code is a new tool which will prove useful for many years to come. Computer codes which provide new capabilities are used for ten to twenty years (or more) and provide a research platform for many future graduate and undergraduate students. As an example, over a period of ten years previous codes have been used by about 20 undergraduate students and 3-5 graduate students.

Impact on Society –

The impact of this project on graduate students, undergraduate students, and K-12 students will impact the overall science literacy of the voting public, making an impact beyond the bounds of science and engineering alone.

New Technology --

None.

Baylor - The ground unit analog PK-4 BU is now in service at Baylor University. Testing and initial calibration is complete as are all christening tests. The instrument tube is identical to the one currently in use at the University of Iowa (John Goree) and as close to the tube employed by European Space Agency/German Aerospace Center (ESA/DLR) in their ground model as possible. All ancillary equipment (power supply, cameras, etc.) is either identical to equipment used on the ISS PK-4 or as close as possible given financial constraints. Analysis of new PK-4 BU data is underway as is comparison to data produced by numerical (CASPER BU) and analytical (University Of California, San Diego-UCSD) models. Certification runs against recent ISS PK-4 data is also underway. The PK-4 BU device has now been used by both the Baylor team, the Auburn team, and the Wittenberg team. Auburn personnel have been on campus to use the device and Auburn / Wittenberg personnel are scheduled to spend time on the device in April, 2019.
Numerical (Baylor) - An entirely new code has been written and tested to model the ion-dust interactions within the PK-4 BU analog device. The code is written in C++ with CUDA to make use of the parallel processing power of Graphical Processing Units (GPUs). The code makes use of boundary conditions for the plasma provided by a PIC-MCC code used to model the PK-4 DC discharge. The IonWake simulation is a MD simulation of ion and dust dynamics, each resolved on their own timescale. The charging of the dust is calculated self-consistently within the ion wake. The dynamics of the ion includes calculations of ion-ion accelerations, ion-dust accelerations, ion-neutral collisions, ions insertions (assuming a shifted Maxwellian velocity distribution incident on the simulation boundary), and ion deletions due to leaving the simulation boundary or impacting the dust particle. The dust dynamics includes calculations of charge, ion-dust accelerations, dust-dust accelerations, external electric fields present in the DC plasma, additional confining electric fields, gas drag, and a thermal gas bath.

The code has been tested for configurations of dust strings under a variety of gas pressures and ion flow velocities, set by the magnitude of the current applied to the polarity switching in the PK-4 experiment. Numerical (Institute for Solid State Physics and Optics, Wigner Research Centre for Physics) - A 2D, cylindrical, particle in cell code including Monte Carlo collisions (PIC/MCC) has been developed to model gas discharge simulations for parameters matching the PK-4 operating regime and experimental data from the PK-4 BU. The current version of the code is capable of accurately describing the gas-phase processes and relevant surface processes on the level of elementary electron-atom and ion-atom collisions. The simulation includes the collisional cross section set published by Biagi (v7.1), which includes electron–neon elastic scattering, ionization and excitation to 9 different (including 2 metastable) energy levels. Ion–atom collisions include isotropic elastic and symmetric charge transfer collisions. Metastable density distribution calculation and Penning ionization are included in the model. Surface processes include electron reflection and ion induced secondary electron emission from the electrodes and surface charging on the glass wall. The results of the simulations provide quantitative information of the charged particle densities, fluxes, electric field, and energy distributions as well as distribution of specific collisional events, like the ionization and excitation source functions. Implementation of this model on GPU accelerated architecture turned out to be of key importance to achieve feasible simulation times.

Russian Academy of Sciences - The influence of a dust cloud on the structure of the positive column of a direct current gas discharge in a cylindrical glass tube under microgravity conditions has been studied both experimentally and numerically [Zobnin et al., Phys. Plasmas 2018]. The discharge was produced in neon at 60 Pa in a glass tube with a diameter of 30mm at a discharge current 1 mA. Spherical monodisperse melamine formaldehyde dust particles with a diameter of 6.86 mu m were injected into the positive column and formed there a uniform dust cloud with a maximum diameter of 14.4 mm. The shape of the cloud and the dust particle number density were measured. The cloud was stationary in the radial direction and slowly drifted in the axial direction. A numerical simulation of the discharge was performed using the 2D (quasi-3D) nonlocal self-consistent kinetic model of a longitudinally inhomogeneous axially symmetric positive column, which was supplemented by a program module performing a self-consistent calculation of dust particle charges, the plasma recombination rate on dust particles, and ion scattering on dust particles. The results of the simulation were compared with experimental data. Within the study of the conditions for the dissipative instability onset in an open particle system with chain-like structure, the problem of the stability of a system consisting of two interacting particles (as a basic form of a chain), which is collinear to the electric field vector was solved [Lisina et al., J. Phys.: Conf. Ser. 2019]. For the model of a wake potential of a point charged particle suspended in a weakly collisional plasma with a directed ion flux the structural stability conditions of the vertical configuration of two interacting particles were obtained depending on the Mach number, the plasma screening parameter, the particle charge ratio, and the ratio

Task Progress:

of the gravity force to the Coulomb interaction force. The oscillatory process in the dusty plasma in the glow discharge stratum was studied experimentally [Petrov et al., J. Exp. Theor. Physics 2018]. The motion of dust particles in an electric field of solitons has been analyzed. These experimental results have been interpreted theoretically using the MHD model of the plasma. The experimental and theoretical investigations of the formation of dust particle clouds in the stratified positive column of a DC glow discharge in helium were performed [Fedoseev et al., Phys. Plasmas 2018]. The size and shape of a dust cloud that levitated in the strong electric field of the striations in a vertically oriented discharge tube were measured under different helium pressures. Axial electric field strength was also experimentally estimated. A model for radial distributions of all dusty plasma parameters of the positive column of the dc glow discharge was developed to describe the obtained experimental results. The model is based on the solution of a non-local Boltzmann equation for an electron energy distribution function, drift-diffusion equations for ions and dust particles, and a Poisson equation for a self-consistent radial electric field. The conditions for the amplitude instability onset in a two-dimensional Yukawa cluster system consisting of seven particles were investigated for the case of sufficiently large thermal particle displacements [Lisina et al., Phys. Rev. E 2019]. We considered an analytical approach to derive the criterion for such instability onset, which is based on searching an energetically favorable inflection point of the system potential energy. We used numerical simulations to study the structural and dynamical properties of the system near the critical point of the amplitude instability. The obtained results were compared to the criteria for melting of the “infinite” two-dimensional system.

DLR – Discussions with DLR continue to occur on a regular basis on (1) verification of BU PK-4 hardware against the DLR PK-4 ground unit and (2) producing the Baylor portion of the Auburn / Baylor / Wittenberg proposal for the upcoming PK-4 Campaign #7 experimental campaign. This experimental proposal has been submitted to DLR and will be discussed at the FST meeting in Munich in February which the PI will attend. With respect to past PK-4 mechanical (i.e., gas leak) issues, the new “Experimental Interface” as been installed in the PK-4 FM (Flight Model) and successfully tested in a dedicated commissioning campaign. Additionally, an extensive search of available ISS PK-4 experimental data has been conducted (with the help of DLR personnel and CASPER personnel) to look for conditions conducive to the formation of particle strings. More recently, an entirely new set of experiments on this subject was flown in Campaign #5 by DLR and that data has also been discussed with the CASPER PI. The ongoing analysis described above by CASPER (Baylor and DLR) personnel has identified (1) various regimes within the PK-4 where particle strings can form, and (2) the coexistence of strings comprised of particles of varying sizes. Both of these are now being verified on the PK-4 BU device.

Specific Objectives Accomplished –

Baylor - Objective 1: Establishing the plasma parameters required for the formation and stability of dust particle chains. The ground unit analog at Baylor (PK-4 BU) is now producing data which is being compared to data produced by numerical (CASPER BU) and analytical (UCSD) models as well as PK-4 Ground Unit data (DLR), PK-4 Parabolic data (DLR and ISS PK-4 data (DLR). An initial set of plasma parameters required to produce stable dust particle chains has now been identified.

Baylor - Objective 1: Establishing the plasma parameters required for the formation and stability of dust particle chains. The MD model has been used to test the chain formation under a variety of ion flow speeds (set by the electric field due to the polarity switching), comparing the ion wake field within the chain as well as the order of the chain structure measured through the correlation function, g(r). These have also been compared with theoretical results based on particle interactions assuming a dipole or quadrupole field for the effective electrorheological dust interaction.

Numerical (Institute for Solid State Physics and Optics, Wigner Research Centre for Physics) - The objective of the PIC/MCC simulation was to quantify the local plasma parameters at the position of the dust particles in the experiment. Information on these parameters is of great importance for further modeling the dust grain charging and inter-grain interactions, as well as for the computation of the confinement forces responsible for the global structure and motion of the dust cloud.

High-speed imaging and photometric experiments are serving as validation of the model and have already potentially shown non-expected phenomena needing further investigation. Russian Academy of Sciences – Over the past year, the development of a new method for determining wake-mediated forces of interparticle interactions in complex plasmas has progressed. Additionally, investigation of the amplitude instability in a chain-like system of dust particles suspended in a plasma with flowing ions has been conducted.

Significant Results –

Baylor - A series of multi-chain structures have been produced and trapped within the PK-4 BU using a DC field with varying polarization switching frequencies and duty cycles. The motion of the particles has been tracked and their positions and velocities recorded for each continuous frame.

Numerical (Baylor and Wigner Institute) -To date, the simulations have revealed that the apparently quiet positive column regime of the neon discharge is actually governed by fast, moving, periodic ionization waves. Further, during the rapid polarity switching process, a normal operation mode in the PK-4 experiment designed to give fine control over the global motion of the dust cloud, an even more violent transient wave occurs. During both periodic and transient waves, large amplitude peaks appear in the local plasma density, electron and ion energies and fluxes, axial electric field, which may be candidates for explaining the observed dust particle chain formation.

High-speed imaging experiments featuring up to 50,000 frames per second have been conducted and fully confirm the presence of these periodic ionization waves. Systematic parametric studies to quantify the gas pressure and electric current dependence on the wave propagation velocities have also been conducted, showing them to be in the range of several 100 m/s. Applying polarity switching to the DC neon discharge and performing synchronized two channel fast photometric measurement we have also verified the presence of a fast transient wave with a phase velocity in the range of 30,000 m/s, in agreement with the PIC/MCC simulation data.

Russian Academy of Sciences - It was found that in the presence of a dust cloud in the structure of the positive column of a DC neon discharge in a cylindrical glass tube under microgravity conditions, the intensity of the neon spectral line with a wavelength by 585.25 nm emitted by the discharge plasma increased by 2.3 times and 2 striations appeared on the anode side of the cloud. It was demonstrated that for the best agreement between simulated and experimental data, it is necessary to take into account the reflection of electrons from the dust particle surface in order to correctly describe the recombination rate in the cloud, its radial stability, and the dust particle charges [Zobnin et al., Phys. Plasmas 2018]. For

the model of wake potential of a point charged particle suspended in a weakly collisional plasma with a directed ion flux the structural stability conditions of the vertical configuration of two interacting particles were obtained depending on the Mach number, the plasma screening parameter, the particle charge ratio, and the ratio of the gravity force to the Coulomb interaction force [Lisina et al., J. Phys.: Conf. Ser. 2019]. The directed ion flow and the wake interaction between dust particles have a dual effect on the stability of vertical chain structure of the system of interacting particles. On the one hand, it is precisely the presence of directed ion flux in the area where dust particles are suspended that causes vertical ordering of dust particles. On the other hand, nonreciprocal interaction with a positively charged ion cloud provides a very efficient mechanism for converting the energy of the flowing ions into the kinetic energy of dust particles, which can lead to another kind of instability onset – the amplitude instability. The problem of the development of amplitude instability in an open system of dust particles with chain-like structure immersed in asymmetric nonequilibrium medium is an interesting matter for future research. The obtained conditions for the dissipative instability onset can be useful for experimental analysis particularly for estimating the electric fields in a DC discharge.

We substantiated theoretically that the dust–acoustic instability, which plays the key role in the oscillatory process, involves the excitation and evolution of a dust–acoustic soliton [Petrov et al., J. Exp. Theor. Physics 2018]. In our opinion, the energy source (pump) for the oscillatory process is the ion drag force that leads to the evolution of nonlinear dust–acoustic waves based on small dust density fluctuations. We showed experimentally that solitons can lead to a large-scale unidirectional transport of charged particles. The results of this study contribute to the development of the physics of nonlinear waves and can also be useful in analysis of transport processes in plasmas.

The experimental and calculated results showed that the size of the dust cloud decreases with the increase in the gas pressure, and the axial electric field strength pressure dependencies have minima [Fedoseev et al., Phys. Plasmas 2018]. At low gas pressures, the dust particle charge number density increases and exceeds the electron density that strongly influences the plasma parameters.

We showed that there is a proportionality between melting curve for an “infinite” two-dimensional system with the hexagonal lattice and analytical curve of critical points for the seven-particle Yukawa cluster [Lisina et al., Phys. Rev. E 2019].

Key Outcomes / Other Accomplishments –

Institute for Solid State Physics and Optics, Wigner Research Centre for Physics - Initial results predict the presence of strong ionization waves in the stationary discharge and the appearance of an ionization tsunami during polarity switching as it is applied in the PK-4 experiment. This has now been confirmed experimentally using the PK-4 BU. The computed plasma parameters are in rough agreement with those observed experimentally and serve as input for dust grain charging models and calculation of interparticle interactions.

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